Automated Segmentation of Solar Filaments

Inter IIT Tech Meet 15.0 Prepathon – Computer Vision Problem Statement

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Abstract

Solar filaments are elongated structures of relatively cool, dense plasma suspended in the solar corona and visible in Hydrogen-alpha ($H\alpha$) observations as dark absorption features. Their eruption is associated with major space-weather phenomena, including coronal mass ejections and solar flares. Automated and pixel-precise segmentation of these structures is therefore an important computer-vision problem for large-scale solar observation analysis.

This report presents an end-to-end segmentation pipeline developed for Task A, Automated Segmentation of Solar Filaments, using the MAGFiLO dataset of native-resolution 2048×2048 H-alpha observations. The system addresses three principal challenges: preservation of ultra-thin filament barbs, rejection of visually similar solar-background structures, and maintenance of coherent filament topology. The proposed solution combines native-resolution biased cropping, a MORDEN-inspired U-Net decoder, transfer learning with an ImageNet-pretrained ResNet34 encoder, one-channel adaptation for grayscale H-alpha imagery, balanced Tversky loss, dihedral test-time augmentation, and lightweight connected-component post-processing.

The final configuration, evaluated natively on the 2048×2048 full-disk validation set, achieved a Mean IoU of 0.4722, global pixel Precision of 0.7215, global pixel Recall of 0.6057, Hit Rate of 0.9075, Miss Rate of 0.0925, AP@IoU 0 of 0.6928, and Multi-scale IoU of 0.4729. Full-disk inference required approximately 0.31 seconds per image on an NVIDIA RTX 4050 with 6 GB VRAM. The resulting pipeline prioritizes structural detection and computational efficiency while operating directly on native-resolution solar observations.

1 Introduction & Data Strategy

1.1 Problem Formulation

The objective of Task A is to produce automated, pixel-precise semantic segmentation masks for solar filaments in full-disk ground-based $H\alpha$ observations. In these observations, filaments appear as dark absorption structures against the brighter solar disk. The segmentation problem is substantially more difficult than conventional binary segmentation because the foreground structures occupy only a very small fraction of the image and contain thin, elongated branches that are easily destroyed by aggressive resizing.

The principal engineering challenges are:

1. **Barb preservation:** ultra-thin, thread-like structures branching from the principal filament spine must be retained rather than eroded during preprocessing or decoding.
2. **Background rejection:** the model must distinguish genuine filament plasma from dark active regions, sunspots, chromospheric fibrils, quiet-Sun granulation, and other visually similar structures.
3. **Structural coherence:** filament predictions should form coherent connected entities rather than disconnected collections of isolated foreground pixels.

The MAGFiLO dataset contains native 2048×2048 single-channel 8-bit grayscale observations together with polygonal contour annotations and associated filament metadata. The experimental pipeline was designed around the preservation of the native physical scale of these images. The supplementary submission package also includes the file `solarfragmentsCV` as part of the submission package reference.

1.2 Dataset Triage and Annotation Engineering

The initial dataset inspection revealed an annotation indexing discrepancy. The raw directory contained 707 physical training images, whereas the COCO-style annotation file contained 1,154 image records. The discrepancy originated from repeated image metadata associated with different filament instances rather than a strict one-to-one correspondence between annotation records and physical files.

Instead of modifying the source annotations or constructing additional persistent manifest files, the discrepancy was resolved in memory. An image dictionary maps image identifiers to file names, while annotations are grouped using Python's `dict.setdefault` operation:

```
image_dict = {img["id"]: img["file_name"] for img in coco["images"]}
ann_dict = {}

for ann in coco["annotations"]:
    ann_dict.setdefault(ann["image_id"], []).append(ann)
```

This grouping operation permits multiple polygonal segments belonging to the same physical image to be merged during processing. Consequently, fragmented annotation records are handled without altering the underlying source data.

1.3 Extreme Foreground Sparsity

Rasterization of the annotated polygons demonstrated severe class imbalance. The mean foreground coverage was only 0.3883%, corresponding to approximately 99.61% background dominance. The median foreground coverage was 0.3298%, with a minimum of 0.0078% and a maximum of 1.7280%. Across the evaluated images, the annotated foreground represented 18,795,931 pixels out of more than 4.8×10^9 evaluated pixels.

This imbalance makes pixel accuracy an unsuitable optimization and evaluation target. A degenerate model predicting background everywhere could exceed 99.6% pixel accuracy while producing an IoU of zero. The development process therefore emphasized overlap, detection, and boundary-sensitive metrics instead of raw accuracy.

1.4 Native-Resolution Biased Cropping

A central preprocessing decision was to avoid resizing the entire 2048×2048 solar disk to 512×512. Such a transformation represents a four-fold spatial decimation in each dimension. Since many filament barbs are only approximately 1–3 pixels wide at native resolution, downsampling can remove these structures before the segmentation network receives the input.

Training therefore uses native-resolution 512×512 crops extracted directly from the 2048×2048 observations. Sampling is deliberately biased:

- With probability 50%, an annotated filament is selected, its polygon centroid is calculated as

$$(\bar{x}, \bar{y}) = \left(\frac{1}{N} \sum_{i=1}^{N} x_i, \frac{1}{N} \sum_{i=1}^{N} y_i \right),$$

and the crop is centered approximately on this location, subject to image-boundary clipping.

- With probability 50%, a crop is sampled uniformly across the solar disk. These background-oriented samples expose the network to quiet-Sun granulation, limb artifacts, empty photosphere, and non-filament dark structures.

This strategy deliberately balances two competing requirements: sufficient exposure to rare foreground structures and sufficient exposure to the difficult background distribution.

1.5 Data Augmentation and Validation Isolation

Solar filaments do not possess a canonical image-space orientation. Horizontal and vertical reflections can therefore be used as geometric augmentations without imposing an artificial preferred direction. The transformations were implemented directly inside the PyTorch dataset pipeline using tensor operations.

During training, horizontal and vertical reflections are independently sampled with probability 0.5. Validation images remain unaugmented and are evaluated at their native spatial resolution to maintain benchmark reproducibility.

The dataset was stratified using filament frequency per image. Histogram bins at approximately $[3, 7, 12, 18]$ were used to distribute images across the training and validation subsets. The resulting split contained 80% training data and 20% validation data, corresponding to 923 mapped annotation entries across 565 physical training images and 231 mapped entries across 142 physical validation images during the development-stage split.

2 Architecture Methodology

2.1 Baseline: Lean U-Net

The initial baseline was a four-stage U-Net using conventional 3×3 convolutions, Batch Normalization, and ReLU activations. This configuration provided a simple proof of concept but displayed two important limitations.

First, its effective receptive field was insufficient for capturing long filament structures efficiently. Second, when coupled with downsampled inputs, the network lost the fine-scale information required to preserve filament barbs. The baseline consequently exhibited delayed convergence and substantial false-negative behavior.

2.2 MORDEN-Lite Decoder Design

The second architecture incorporated selected high-impact ideas inspired by the MORDEN approach. The principal modifications were:

1. **Expanded convolutional kernels:** decoder convolutions were changed from 3×3 to 5×5 kernels with padding equal to 2. The larger kernels increase the local receptive field and help the decoder trace elongated filament bodies without requiring excessive additional pooling.

2. **Instance Normalization:** `BatchNorm2d` was replaced with `InstanceNorm2d`. H-alpha observations exhibit substantial image-to-image variation in global illumination, limb darkening, and exposure. Instance normalization performs normalization independently for each image and is therefore better aligned with this heterogeneous input distribution.

3. **LeakyReLU activation:** ReLU was replaced with LeakyReLU using a negative slope of 0.01. This preserves a non-zero gradient for negative activations and reduces the risk of inactive units in a setting where the foreground is extremely sparse.

These changes produced a MORDEN-Lite U-Net that was specifically oriented toward the combination of large-scale structure recognition and thin-structure preservation.

2.3 Final Architecture: ResNet34-U-Net

The final system adopted a transfer-learning configuration consisting of a PyTorch U-Net decoder attached to an ImageNet-pretrained ResNet34 encoder. The backbone provides a hierarchy of learned edge and texture features while retaining the U-Net skip connections necessary for accurate spatial reconstruction.

2.3.1 One-Channel Adaptation

Because the input consists of single-channel H-alpha grayscale images, directly replicating the grayscale image into three channels would increase the input representation without adding new information. Instead, the first convolution of ResNet34 was adapted from three input channels to one.

Let W_{RGB} denote the original first-layer convolutional weights. The adapted weights are defined as

$$W_{\mathrm{new}} = \sum_{c=1}^{3} W_{\mathrm{RGB}}[:, c, :, :] \in \mathbb{R}^{64\times 1\times 7\times 7}.$$

This preserves the aggregate response of the pretrained RGB filters while allowing the model to consume single-channel H-alpha observations directly. The approach avoids explicit channel duplication and was compatible with the available 6 GB VRAM budget.

2.3.2 Decoder Construction

The decoder progressively reconstructs the full-resolution segmentation map using transposed convolutions, skip connections, and convolutional decoding blocks. The principal stages are:

- Stage 4: $512 \to 256$ transposed convolution, concatenated with a 256-channel encoder feature map, followed by decoding to 256 channels.
- Stage 3: $256 \to 128$ transposed convolution, concatenated with a 128-channel skip feature, followed by decoding to 128 channels.
- Stage 2: $128 \to 64$ transposed convolution, concatenated with a 64-channel skip feature, followed by decoding to 64 channels.
- Stage 1: $64 \to 64$ transposed convolution, concatenated with the first 64-channel encoder feature map, followed by decoding to 64 channels.
- Final reconstruction: $64 \to 32$ transposed convolution, followed by decoding to 16 channels and a 1×1 convolution producing the single-channel segmentation logit.

The decoder retains Instance Normalization and LeakyReLU throughout the reconstruction stages.

3 Optimization & Post-Processing

3.1 Loss-Function Evolution

The optimization strategy evolved through three principal loss formulations.

3.1.1 BCE + Dice Baseline

The initial objective combined binary cross-entropy with Dice loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum \left[y \log(\sigma(p)) + (1-y)\log(1-\sigma(p)) \right],$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum p \cdot y + \epsilon}{\sum p + \sum y + \epsilon},$$

$$\mathcal{L}_{\text{Combo}} = 0.5\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{Dice}}.$$

Although this combination is widely applicable to binary segmentation, the severe foreground imbalance caused the BCE contribution to become dominated by background pixels. This limited the quality of gradients associated with sparse filament structures.

3.1.2 Focal-Dice Loss

The next iteration incorporated focal weighting so that easy examples, particularly abundant background pixels, contributed less strongly to the optimization. The probability assigned to the ground-truth class is

$$p_t = p\,y + (1-p)(1-y),$$

with focal loss expressed as

$$\mathcal{L}_{\text{Focal}} = \alpha(1-p_t)^{\gamma}\mathcal{L}_{\text{BCE}}.$$

The adopted focal parameters were $\alpha = 0.66$ and $\gamma = 2.0$, combined with Dice loss using a balance weight of $w = 0.5$.

3.1.3 Balanced Tversky Loss

Transfer learning substantially improved detection recall but also increased false-positive responses in quiet-Sun granulation and active regions. Dice loss treats false positives and false negatives symmetrically, which is not ideal when false alarms are a dominant failure mode.

The Tversky index was therefore adopted:

$$\text{TI} = \frac{\sum p \cdot y + \epsilon}{\sum p \cdot y + \alpha\sum((1-p)\cdot y) + \beta\sum(p\cdot(1-y)) + \epsilon},$$

with

$$\mathcal{L}_{\text{Tversky}} = 1 - \text{TI}.$$

The coefficients control the relative penalties assigned to false negatives and false positives. Increasing β relative to α increases the penalty on false-positive predictions.

An initial configuration of $\alpha = 0.3$ and $\beta = 0.7$ increased Precision to 0.6822 but reduced Recall to 0.5496. The final configuration used

$$\boxed{\alpha = 0.4, \qquad \beta = 0.6}$$

which provided a more balanced precision-recall operating point and produced the final IoU improvement.

3.2 Native Full-Disk Inference

The final U-Net is fully convolutional and therefore supports arbitrary spatial dimensions satisfying the architectural stride constraints. The network requires spatial dimensions compatible with the cumulative downsampling factor of $2^5 = 32$.

At inference time, the complete 2048×2048 image is processed as a single tensor rather than being tiled into overlapping patches. This avoids seam artifacts and boundary inconsistencies introduced by tile stitching. Gradient storage is disabled during evaluation, allowing the complete solar disk to be processed within the available GPU memory.

3.3 Test-Time Augmentation

To reduce directional bias, the final inference pipeline uses a three-pass dihedral augmentation ensemble consisting of the original image, a horizontally flipped image, and a vertically flipped image. Predictions from the transformed images are restored to the original coordinate system before averaging:

$$P_{\text{ensemble}} = \frac{1}{3}\left[\sigma(f(I)) + \text{flip}_H\left(\sigma(f(\text{flip}_H(I)))\right) + \text{flip}_V\left(\sigma(f(\text{flip}_V(I)))\right)\right].$$

The resulting probability map is thresholded at $p > 0.5$. TTA adds inference computations but does not introduce additional trainable parameters. In practice, the ensemble smooths thin structures and reduces orientation-specific prediction artifacts.

3.4 Connected-Component Post-Processing

Dense Conditional Random Fields were considered as a post-processing mechanism but rejected because of their computational cost and observed failure characteristics. A reported runtime of approximately 6.5 seconds per 2048×2048 image is incompatible with the target low-latency pipeline. In addition, pixel-gradient-driven refinement can lock onto non-filament dark structures such as sunspots.

The final pipeline uses OpenCV's optimized `connectedComponentsWithStats` with eight-way connectivity. Components smaller than 64 pixels are removed:

$$\text{Mask}_{\text{clean}} = \{C_k \in C \mid \text{Area}(C_k) \geq 64\}.$$

This simple geometric filter removes isolated background speckles while retaining sufficiently large coherent filament components. The resulting post-processing stage is computationally lightweight and contributes to the measured end-to-end latency of approximately 0.31 seconds per full disk.

4 Final Results

4.1 Chronological Experimental Progression

The engineering iterations demonstrate the effect of progressively addressing data scale, receptive field, transfer learning, and precision-recall imbalance.

Table 1: Chronological development of the segmentation pipeline.

Phase	Model Setup	Resolution	IoU	Prec.	Recall	Key Milestone
1	Plain U-Net (3×3, BatchNorm, BCE+Dice)	512×512 resized	0.3139	–	–	Baseline proof of concept
2	MORDEN-Lite (5×5, InstanceNorm, Focal-Dice)	512×512 biased crops	0.5157	–	–	Improved barb preservation through native patches
3	MORDEN-Lite + OpenCV area filter	2048×2048 full disk	0.3815	0.5483	0.5931	First full-disk evaluation
4	ResNet34-U-Net + collapsed Conv1 + Focal-Dice	2048×2048 full disk	0.3880	0.4631	0.7612	High recall; precision degraded on active regions
5	ResNet34-U-Net + Tversky ($\alpha = 0.3, \beta = 0.7$)	2048×2048 full disk	0.4179	0.6822	0.5496	Suppressed background false alarms
6	ResNet34-U-Net + Tversky ($\alpha = 0.4, \beta = 0.6$)	2048×2048 full disk	0.4700	0.6846	0.6229	Improved precision-recall balance
Final	**ResNet34-U-Net + Balanced Tversky + TTA**	2048×2048 **full disk**	**0.4722**	**0.7215**	**0.6057**	**Official competition submission configuration**

4.2 Official Validation Metrics

The final configuration was evaluated natively on 231 full-disk validation images using the competition criteria. For binary segmentation,

$$\text{IoU} = \frac{|P \cap GT|}{|P \cup GT|},$$

while object-level Hit Rate is defined as

$$\text{Hit Rate} = \frac{\text{Detected Ground Truth Objects}}{\text{Total Ground Truth Objects}}.$$

The official results are summarized in Table 2.

Table 2: Final official evaluation metrics on the full-disk validation set.

Metric	Result
Mean IoU	**0.4722**
Global Pixel Precision	**0.7215**
Global Pixel Recall	**0.6057**
Hit Rate	**0.9075**
Miss Rate	**0.0925**
AP@IoU 0	**0.6928**
Multi-scale IoU (MIoU)	**0.4729**
Inference Latency	~ 0.31 **s / full disk**
Throughput	~ 3.23 **images/s**

The Hit Rate corresponds to successful detection of 1,482 out of 1,633 ground-truth filament objects, leaving 151 missed objects. The AP@IoU 0 value of 0.6928 corresponds to 1,838 valid

detections among 2,653 predicted components. The MIoU of 0.4729 was obtained by evaluating the segmentation across a scale pyramid containing 1×, 0.5×, and 0.25× image scales.

4.3 Diagnostic Ablations and Failure Modes

4.3.1 Scale-Variance Failure

An experiment evaluated the model at 1024×1024 rather than native 2048×2048 resolution. Mean IoU decreased to 0.3523, while Recall fell from 0.6229 to 0.4456 in the corresponding comparison.

This degradation is consistent with the scale relationship between training and inference. The network was trained on native-resolution 512×512 crops extracted from the 2048×2048 imagery. Consequently, the learned convolutional filters became calibrated to the physical filament widths present at native scale. Halving the full-disk resolution correspondingly reduced the apparent filament thickness and caused thin structures to become more difficult to activate and reconstruct.

4.3.2 Pseudo-Labeling Failure

High-confidence pseudo-labeling was also investigated as a mechanism for increasing training diversity on unlabeled frames. Predictions above $p > 0.85$ were considered as candidate pseudo-labels.

The experiment was rejected because the approach introduced confirmation bias. Dark quiet-Sun fibrils and other non-filament structures could be incorporated as high-confidence false-positive labels, after which subsequent training reinforced these errors. The procedure therefore destabilized precision rather than providing a reliable improvement.

4.3.3 Hit Rate Versus Mean IoU

The final system achieved a Hit Rate of 90.75%, indicating that 1,482 of 1,633 ground-truth filament objects were successfully detected. At the same time, Mean IoU remained moderate at 0.4722.

The difference between these measurements is explained by the extreme geometry and sparsity of the target structures. Filaments occupy less than approximately 0.4% of image area on average and can consist of structures only a few pixels wide. A boundary displacement of even 1–2 pixels can therefore produce a large relative reduction in pixel-level overlap while the filament itself remains correctly localized as an object. Thus, object-level detection and pixel-level boundary agreement measure different aspects of segmentation quality.

4.4 Comparison with the Reference MORDEN Pipeline

The reported reference systems in the MORDEN literature operate with different datasets, spatial scales, hardware, and post-processing configurations. Consequently, their numerical metrics should not be interpreted as directly equivalent benchmarks.

The reference MORDEN configuration reported a Mean IoU of 0.6700 on the HAS dataset, while the reported MORDEN SOTA configuration on the AHAS dataset achieved a Mean IoU of 0.8120. Their configurations use substantially different computational resources and post-processing pipelines, including DenseCRF and DBSCAN. In contrast, the present system uses MAGFiLO, native 2048×2048 evaluation, a ResNet34-U-Net backbone adapted to one-channel imagery, and connected-component filtering.

The apparent performance gap in Mean IoU must be considered in the context of the differing evaluation setup. The present system is evaluated on native, uncompressed 2048×2048 images, where filaments remain extremely sparse and thin in pixel space. In contrast, the reference

Table 3: System-level comparison with the reported MORDEN configurations.

Metric / Parameter	**MORDEN Paper (HAS)**	**MORDEN SOTA (AHAS)**	**Proposed MAG-FiLO Pipeline**
Model Backbone	Custom Multi-Head ConvNet	MORDEN + APP + PAG	ResNet34-U-Net (1-channel adapted)
Post-Processing	DenseCRF + DB-SCAN	DenseCRF + DB-SCAN	Connected Components, area ≥ 64
Evaluation Scale	1024×1024	2048×2048 or 1024^2	Native 2048×2048 full disk
Hardware	NVIDIA RTX H100	NVIDIA RTX H100	NVIDIA RTX 4050, 6 GB VRAM
Inference Latency	~ 0.17 s (1024^2)	~ 6.5 s (2048^2)	~ 0.31 s (2048^2)
Mean IoU	0.6700	0.8120	0.4722
F_1 Score	0.8090	0.8950	0.6523
Precision	–	–	0.7215
Hit Rate	–	–	0.9075

experiments involve different datasets and image scales, making direct numerical comparison imperfect.

A more directly comparable engineering characteristic is computational efficiency. Replacing the iterative DenseCRF operation, reported at approximately 6.5 seconds per full-disk image, with an optimized connected-component filter reduces the post-processing burden substantially. The final pipeline processes a full disk in approximately 0.31 seconds, corresponding to roughly 3.23 images per second.

4.5 Final System Characteristics

The final competition configuration can be summarized as follows:

1. Native 2048×2048 H-alpha imagery is used as the canonical evaluation scale.
2. Training avoids destructive full-image downsampling and instead uses native-resolution 512×512 biased crops.
3. A ResNet34 ImageNet encoder is adapted from RGB to single-channel H-alpha input by collapsing the first convolution weights.
4. The decoder uses MORDEN-inspired 5×5 convolutions, Instance Normalization, and LeakyReLU activations.
5. Balanced Tversky loss with $\alpha = 0.4$ and $\beta = 0.6$ provides explicit control over the false-positive versus false-negative trade-off.
6. Full-disk inference is performed in a fully convolutional manner without tile stitching.
7. D4-style horizontal and vertical flip TTA improves prediction consistency without increasing parameter count.
8. A connected-component area filter removes isolated foreground speckles at substantially lower computational cost than DenseCRF-based refinement.

4.6 Conclusion

The final segmentation system demonstrates that robust solar-filament segmentation depends not only on model architecture but also on preserving the physical scale of the targets throughout the data pipeline. The most significant engineering decisions were the rejection of destructive full-disk downsampling, the use of centroid-biased native-resolution crops, adaptation of a pretrained ResNet34 encoder to single-channel imagery, and explicit optimization of the precision-recall trade-off through balanced Tversky loss.

On the official full-disk validation set, the pipeline achieved Mean IoU of 0.4722, Precision of 0.7215, Recall of 0.6057, Hit Rate of 0.9075, Miss Rate of 0.0925, AP@IoU 0 of 0.6928, and MIoU of 0.4729, with approximately 0.31 seconds of inference latency per 2048×2048 image on an NVIDIA RTX 4050 with 6 GB VRAM.

The results indicate that the system is particularly effective at locating filament objects while maintaining practical inference speed on constrained hardware. The remaining gap between high object-level Hit Rate and moderate pixel-level IoU is primarily associated with the extreme sparsity and fine-scale geometry of filament boundaries, highlighting boundary localization as the principal opportunity for future improvement.

References

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